

SOC Automation Home Lab

Implementation Plan

Structure: 30-60-90 Day Actionable Roadmap

Target Stack: Wazuh SIEM, Kali Linux, Automated Threat Simulation, File Integrity Monitoring

Phase 1: Days 1–30 — Core Engineering & Foundational Visibility

Focus on establishing isolated host networks, standing up the central collector nodes, and achieving initial log capture visibility.

Days 1–3: The Security Brain & Threat Validation (Complete

Import the central Wazuh SIEM appliance, fix persistent networking loopback configurations via hard-coded overrides, register the Kali Linux attacker node on the private Host-Only network line, and validate end-to-end routing using rapid packet exchanges. Trigger and catch the initial level-10 brute-force security alert patterns (Rule 5712) inside the live dashboard indexers.

Days 4–10: External Agent Deployment & Endpoint Ingestion

Provision target victim operational instances (Ubuntu Server / Windows Client). Deploy the native lightweight Wazuh monitoring daemon onto external assets, authenticate the client keys against the manager core, and verify remote system health and log streaming ingestion inside the unified Discover panel.

Days 11–20: File Integrity Monitoring (FIM) Policies

Configure directory watch triggers within the client configuration runtime matrices. Engineer absolute monitoring loops across critical system storage spaces (e.g., /etc system configurations or Windows System32 hives) to detect real-time tampering, permission alterations, and malicious file drops.

Days 21–30: Local Firewalls & Baseline Customization

Lock down asset parameters by establishing basic local packet inspection filtering rules. Strip out noisy log variables from the dashboard ingestion engines to optimize query performance and document the active system state inside the core repository logs.

Phase 2: Days 31–60 — Automated Simulations & Detection Engineering

Focus on transitioning from basic log collection to proactive vulnerability scanning and custom alert handling frameworks.

Days 31–40: Ingesting Vulnerability Matrix Scans

Enable the native Wazuh Vulnerability Detector modules. Correlate local system package databases against real-time global CVE asset catalogs to automatically identify critical unpatched flaws across all active target endpoints.

Days 41–50: Custom Rule Engineering & Correlation Logic

Write customized XML alert decoder rules inside the central SIEM rule managers. Teach the monitor engine to isolate nuanced patterns—such as out-of-bounds user privilege elevation behaviors—and upgrade general warning events into critical high-level alerts.

Days 51–60: Automated Threat Scripting

Leverage Kali Linux to trigger advanced tactical simulations beyond basic brute-force routines. Script mock network sweeps, directory reconnaissance maps, and bad file lookups to ensure custom decoder logic fires immediately under stress conditions.

Phase 3: Days 61–90 — Active Response, MITRE Mapping & Metrics

Focus on implementing automated defensive actions, tracking against global playbooks, and cementing the engineering portfolio.

Days 61–75: Active Response Mitigation Frameworks

Configure automated script execution blocks inside the endpoint agent configurations. Teach the environment to execute rapid local protective containment protocols—such as auto-blocking malicious source IP addresses or severing unauthorized network loops—the second a level-12 rule breach occurs.

Days 76–85: MITRE ATT&CK Framework Dashboard Alignment

Map all active home-lab rule configurations directly against the enterprise MITRE ATT&CK matrix layout matrices. Validate that defensive alerts clearly identify attack methodologies (e.g., Persistence, Privilege Escalation) to demonstrate real-world analysis capability to engineering recruiters.

Days 86–90: Comprehensive Architecture Documentation

Assemble all custom configuration scripts, endpoint onboarding files, and attack capture captures into the master public GitHub directory. Finalize a comprehensive, executive-ready technical portfolio highlighting full active monitoring capability.

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